



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

ANTON PAAR USA, INC.  
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CALIBRATION

Valid To: June 30, 2018

Certificate Number: 2697.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations<sup>1</sup>:

I. Fluid Quantities

| Parameter/Equipment                                              | Range                                                                                                                                                                              | CMC <sup>2,4</sup> (±)                         | Comments                                                                              |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| Viscosity <sup>3</sup> –<br>Kinematic (SVM)<br><br>(20 to 40) °C | < 10 mm <sup>2</sup> /s<br>(10 to 100) mm <sup>2</sup> /s<br>(>100 to 1000) mm <sup>2</sup> /s<br>(>1000 to 10 000) mm <sup>2</sup> /s<br>(> 10 000 to 100 000) mm <sup>2</sup> /s | 0.22 %<br>0.28 %<br>0.36 %<br>0.46 %<br>0.52 % | Internal calibration procedure<br><br>Note: calibration typically performed at 40 °C. |
| Viscosity <sup>3</sup> –<br>Rotational<br><br>20°C<br><br>25°C   | <br><br>(1350 to 1650) mPa·s<br><br>(990 to 1210) mPa·s                                                                                                                            | <br><br>13 mPa·s<br><br>12 mPa·s               | Internal calibration procedure<br><br>Note: calibration typically performed at 20 °C. |

| Parameter/Equipment                                                | Range                           | CMC <sup>2</sup> (±)   | Comments                                                                              |
|--------------------------------------------------------------------|---------------------------------|------------------------|---------------------------------------------------------------------------------------|
| Density with U-Tube Technology <sup>3</sup> –<br><br>(15 to 40) °C | (650 to 1800) kg/m <sup>3</sup> | 0.05 kg/m <sup>3</sup> | Internal calibration procedure<br><br>Note: calibration typically performed at 20 °C. |

<sup>1</sup> This laboratory offers commercial calibration service and field calibration service.

<sup>2</sup> Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of  $k = 2$ . The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

<sup>3</sup> Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

<sup>4</sup> All CMCs are listed in percent of reading of input unless otherwise stated.



# ***Accredited Laboratory***

A2LA has accredited

**ANTON PAAR USA, INC.**

*Ashland, VA*

for technical competence in the field of

**Calibration**

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 1<sup>st</sup> day of August 2016.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt', is shown above a horizontal line.

Senior Director of Quality and Communications  
For the Accreditation Council  
Certificate Number 2697.01  
Valid to June 30, 2018

*For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.*